Independent review of the expanded projective-group theory

Distinct mathematical review

11 October 2026

Decision. The frozen five-application edition is accepted with its stated principal contracts, retained inputs, corrections and ancillary exclusions. No mathematical repair is requested. One bibliography locator should be corrected in a separate edition: Schulte–Weiss (1991), *Chiral polytopes*, has pages 493–516, not 435–516. The bound L043 bibliography, item 35, confirms this. The mistake does not affect the theorem interface.

The review binds author manifest `533612648b17120fada68d10141d4852d9b90b93ccf3bcca4f6163975bcad640`. All author files remain unchanged. The reviewer previously reviewed the three-application E theory and worked on its readable edition and measurements. The new chiral proofs have a distinct author. Earlier accepted primitive-square work is explicitly reused; this review does not assign new historical priority or question credit.

Written mathematics and source correspondence

The entire common paper, all five applications, both complete new primary bodies and all required computations were read. The new primaries are L056 (eight pages) and L043 (the 25-page arXiv 1606.08017v1 edition, 2016). The older three source contracts retain their earlier distinct source review; all their current written proofs were reread. Their 32 theorem/proof environments were independently compared with the pinned predecessors and agree literally. The 56 new and 54 retained source dispositions were checked individually. These are dispositions, not 110 new results or 110 originals.

The common reversal proof uses the actual additive commutator $AB - BA$. The intertwining identities alone do not suffice: the diagonal and Jordan arguments establish invertibility under absolute irreducibility, and generating projective pairs discharge that hypothesis. The rotation adapter applies the result to a and ab, giving the required action on b. It includes the small fields. The source's characteristic-two count in Lemma 4.3 is corrected from q to $q - 1$ because the identity is not an involution.

The pointed construction identifies both affine end groups and their precise diagonal intersection. Thus the condition $-1 \in \langle t \rangle$ is necessary for the stated intersection property. Generation by opposite unipotents is proved separately from the semilinear reversal obstruction. The normalization, field degree and Frobenius orbit arguments justify completeness and $\varphi(k)/d$ counts. In particular, a field inversion can give a regular marking of the full group; it does not force a proper subfield group.

Rank descent uses the full indexed rotation intersection property. The anchored involutions and the even subgroup are identified with their actual indexed subgroups; the index-two assertion is not inferred from relations alone. The natural-subfield envelope has its stated embedding and field-size hypotheses. Label-two exclusions, the directly regular subgroup alternatives, and the rank-five and rank-six arguments then give the required rank bound. The source's more general abstract subfield-intersection assertion is not an unstated premise.

The quadratic construction and finite obligations

The trace normal form requires $d = a^2+b^2-4 \neq 0$. Its determinant calculation establishes irreducibility, and the basis I, A, B, AB proves the pair-conjugacy claim. The extension equations have the explicit inverse parameter $z = UR + VS$, with $z^2 = \Delta$. The zero-discriminant case is included. The assertion that reversal occurs precisely at $\Delta = 0$ retains the nonzero-trace and nondegeneracy hypotheses used in its proof.

The exceptional-end classification uses a complete local rotation census, not the earlier involution census as a substitute. The finite universal quotients exclude the spherical and collision cases. They are exact presentation computations, not a sample of possible quotients. The twelve A_4 collision cases in source Lemma 5.12 include four groups isomorphic to A_5, correcting the printed assertion that all twelve are trivial; the claimed projective exclusion still follows. The exceptional intersections at $p = 11$ and 19 are handled explicitly. The residue $10 \pmod{11}$ omitted from the source introduction is restored, in agreement with its Theorem 11.

The complete earlier primitive-square argument is reproduced, including squarefreeness, disjoint roots, the non-power condition for the retained character estimate, the primitive sieve, the integer bounds and all eleven small-field certificates. Equal and nested odd degrees are included. The translated-subfield proof establishes the exact common unipotent subgroup before proving generation. Both square-root choices and the separate semilinear reversal obstruction are addressed. Finite examples are not used to infer these uniform conclusions.

Six whole programs and one mandatory data file are mathematical premises: the earlier regular census, exceptional amalgam census, exceptional small-field check, small rotation exclusions, integer square bound, and direct polynomial witness verifier with its data. Six other complete programs are optional validation. All twelve ran freshly. The Python child processes retained assertions under an optimized outer driver. GAP's exact algorithms and libraries remain trusted inputs.

An independently written tuple-matrix/permutation program imports no author implementation. It checks 316 trace triples and 264 extensions over fields of orders 3, 5 and 7; all 152 applicable reversal tests; and all 432 local rotation pairs in A_4, S_4, A_5. Separate projective subgroup enumeration checks the critical intersections over orders 11, 19 and 31. It finds the order-ten intersection in the zero-discriminant 353 case at 11, and orders three, three and six in the three 535 cases at 19. Three explicit weakenings show why irreducibility, the nonzero denominator, and the pointed -1 condition cannot simply be omitted. These are bounded diagnostics, not replacements for the written proofs.

Formal, visual and contractual limits

All seven Lean modules compiled freshly with 42 standard-axiom reports. Only nine reports are new: the additive-commutator identities and the scalar quadratic parameterization in both directions. The 33 reused reports preserve their original hypotheses and limited conclusions. Two exact inherited `unnecessarySimpa` warnings are harmless style diagnostics. Absolute irreducibility, finite-field parameter existence, projective classification, geometric realization, character estimates and source-specific finite-census correctness are not fully formalized.

All 61 current PDF pages and all 33 new primary pages were viewed. The layout and source correspondence were inspected using labeled page panels, paired with the complete TeX, program and primary-text reading. No clipping or missing notation was identified. All eight current documents were freshly rendered with per-page extracted-text equality to the frozen edition. This is not a browser test or a claim that reduced page panels detect every possible glyph defect.

The retained outside inputs remain visible at use: subgroup and automorphism classification, Schulte–Weiss rotation realization and reversal, the classical regular-polytope correspondence, finite-field facts, the character bound, Dirichlet's theorem and the Suzuki fixed-point theorem in the older application. The full Schulte–Weiss primary is not locally bound. The L043 all-field quotient lists in Theorems 8 and 10 and the complete atlas tables remain ancillary exclusions; the principal existence and classification assertions are recovered without them. The older L079 seventeen-case semilinear census and the separate L021 question track are not silently added to this contract.

The fresh author replay rejects 15 controls. The independent review checker rejects fourteen further scope, evidence and byte corruptions, including false small-field intersections, erased corrections, omitted certificate data, unknown external costs replaced by zero, and a full-formalization claim. These checks bind the mathematical reading; they do not mechanize it.

No numerical compression factor is evaluated or admitted by this review. The whole expanded common theory is a dependency of the five applications. Printed and raw program representations are alternatives. Optional validation must be included on both sides of a comparison or excluded on both sides; unprovided historical code, external proofs, graphics and runtime are not assigned zero cost.

Reproduction. The accompanying `README.rst` gives the fresh full scientific command and the independent binding/diagnostic command. The JSON records preserve every source disposition, exact hashes, reviewer roles and replay evidence. This acceptance introduces no source-count, question-count, ratio or priority claim.